INTERACTION AS A SCIENTIFIC EXPLANANDUM

Boundary review, minimum ontology, and discipline test

DECISION

Do not create an independent discipline at present.

The evidence supports a focused interdisciplinary research program within HCI, tested against strong baselines from existing fields.

Research report

15 July 2026

Scope: human interaction with computational processes, independent of any particular platform or implementation

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Executive decision

> **Decision: do not create an independent scientific discipline at present.** Human interaction with computational systems is a coherent explanandum, but the candidate mechanisms are already substantially covered by Human-Computer Interaction (HCI), cognitive science and psychology, human factors and cognitive systems engineering, ecological psychology, situated action, distributed cognition, activity theory, learning science, semiotics, information visualization, cybernetics, and computational interaction. The unresolved problem is integration: building and testing a representation-aware, context-conditioned, longitudinal model that connects these existing mechanisms across time scales. That problem justifies a focused interdisciplinary research program inside HCI, not a new discipline.

The conclusion is conditional rather than permanent. A separate discipline would become scientifically justified only if a research program demonstrated all of the following:

- persistent cross-domain phenomena that strong combinations of existing theories cannot explain;
- at least one indispensable and operationalizable mechanism not reducible to established constructs;
- replicated out-of-sample predictive improvement across systems, tasks, modalities, populations, and cultures;
- a cumulative ontology and measurement program that remains stable when particular implementations disappear;
- distinctive methods or evidence standards required by the explanandum rather than by institutional preference.

Current literature does not meet those conditions. It instead shows theoretical pluralism, terminological fragmentation, and weak integration. Those are real scientific problems, but fragmentation alone is not evidence of a new natural kind or a new discipline.

Recommended institutional form

Use **foundational human-computational interaction research** as a descriptive program name, not as a disciplinary claim. Place it within HCI and connect it explicitly to cognitive science, human factors, learning science, and social theory. Avoid introducing a new field name until the proposed benchmark program produces evidence of conceptual irreducibility and incremental explanatory power.

Minimum theoretical contribution

The minimum useful contribution is not a new vocabulary. It is a shared integration contract consisting of:

1. a canonical ontology that distinguishes agents, state, representation, interpretation, action opportunities, actions, transitions, feedback, learned state, context, and evidence;
2. typed equivalence relations that prevent semantic, informational, computational, task, and interaction equivalence from being conflated;
3. a context-complete interaction record that makes behavioral traces interpretable relative to available and perceived alternatives;
4. a coupled, partially observed, longitudinal model that permits existing theories to be compared rather than rhetorically combined;
5. decisive falsification and model-comparison tests against a strong baseline assembled from existing disciplines.

1. Research question and decision standard

The central question is not whether interaction is important, complex, or under-theorized. It is whether interaction with computational systems constitutes an explanandum whose mechanisms cannot be adequately developed within existing disciplines.

A scientific discipline is justified here only if it adds explanatory capacity, not merely a community, journal, curriculum, or label. The decision therefore uses five tests.

1.1 Discipline-necessity tests

Test 1 - Domain coherence. Is there a stable class of phenomena that can be identified independently of particular products and implementations?

Test 2 - Conceptual irreducibility. Does explaining that class require constructs or mechanisms that cannot be expressed adequately in established terms?

Test 3 - Predictive increment. Do the proposed additions improve prediction, retrodiction, intervention, or transfer over a strong composite baseline from existing theories?

Test 4 - Methodological necessity. Does the explanandum require distinctive forms of evidence or method rather than ordinary combinations of experimentation, field study, modeling, simulation, trace analysis, and qualitative inquiry?

Test 5 - Cumulative stability. Can the theory accumulate across technologies, tasks, cultures, and historical periods without repeatedly redefining its basic objects?

The present assessment is:

Test	Current result	Basis
Domain coherence	Partly satisfied	Human-computational interaction can be defined as temporally extended reciprocal causal coupling, independent of a named product.
Conceptual irreducibility	Not satisfied	Every candidate core concept has close counterparts in one or more established fields.
Predictive increment	Not demonstrated	No cross-domain benchmark shows that interaction-specific constructs outperform a principled integration of existing models.
Methodological necessity	Not satisfied	The required methods already exist, although their integration and reporting standards are uneven.
Cumulative stability	Not yet demonstrated	HCI contains enduring laws and models, but the field's use of "interaction" remains conceptually diverse and often underspecified.

The negative conclusion follows mainly from Tests 2 and 3. A coherent domain is not sufficient for a new discipline if its explanatory mechanisms are already available and no incremental success has been shown.

1.2 Existing labels do not establish an autonomous discipline

A nominal precedent exists: the *Journal of Interaction Science* began in 2013, and its revised scope described Interaction Science as a deliberately broad, transdisciplinary umbrella spanning biological and technical parties, multiple methods, and socio-technical design (Bahr & Stary, 2016). This demonstrates an institutional program, not conceptual irreducibility. Its definition depends explicitly on integrating psychological, social, engineering, and design traditions.

The same pattern appears inside HCI. HCI has been described as a science, an inter-discipline, and a field with multiple concepts of interaction rather than a single stable theoretical core (Blackwell, 2015; Reeves, 2015; Hornbæk & Oulasvirta, 2017; Hornbæk et al., 2019). A 2025 synthesis treats theories of interaction - information and control, dialogue, tool use, automation, rationality, and practice - as a core part of HCI rather than as a separate discipline (Hornbæk, Kristensson, & Oulasvirta, 2025). Computational interaction and computational rationality likewise present formal explanatory programs within HCI (Oulasvirta et al., 2018; Oulasvirta, Jokinen, & Howes, 2022).

A field label therefore exists in a weak sociological sense, but no reviewed program establishes the stronger scientific claim required here: a stable, non-redundant ontology and mechanism set that cannot be housed in existing disciplines.

2. Scope and core explanandum

2.1 Scope

The scope is **human interaction with computational processes**, not the design of any particular interface and not every relation between humans and technology. A computational process may be deterministic, probabilistic, adaptive, distributed, autonomous, embedded, or materially embodied. The theory must survive replacement of current devices, software platforms, visual conventions, and organizations.

This scope includes:

- moment-to-moment perception-action coupling;
- human and computational initiative, including asynchronous action;
- learning, adaptation, habit, skill, and transfer across episodes;
- representations and their interpretation;
- physical, social, organizational, linguistic, and cultural mediation;
- breakdown, ambiguity, negotiation, and repair;
- individual and collective activity when computational artifacts participate in the cognitive or organizational system.

It excludes, unless linked to reciprocal coupling:

- one-way exposure to static information;
- general cognition with no computational process in the causal loop;
- product evaluation whose claims do not generalize beyond the implementation;
- normative design recommendations presented as explanations.

2.2 Working definition of interaction

Human-computational interaction is a temporally extended reciprocal causal coupling in which human action or state conditions subsequent computational states or actions, and computational action or state conditions subsequent human observations, action opportunities, actions, or learned state.

Reciprocal does not mean equal, synchronous, conversational, or beneficial. Coupling may be intermittent, delayed, opaque, probabilistic, socially mediated, or distributed across people and artifacts. A single event such as a click is not by itself an interaction; it is one observation within a trajectory whose meaning depends on prior state, alternatives, consequences, and interpretation.

2.3 Explanatory target

A general theory should explain or predict the distribution of interaction trajectories and outcomes given:

- the current joint human-computational-environmental state;
- goals, tasks, motives, and institutional objects;
- what actions are feasible, signaled, and believed possible;
- the representations through which state and action possibilities are made available;
- human capabilities, beliefs, skills, expectations, and prior learning;
- computational transition and action policies;
- physical, social, cultural, linguistic, and organizational context;
- feedback, latency, uncertainty, and consequences.

This is a coherent target. The literature review below asks whether it is already decomposed by established disciplines.

3. Method and evidential basis

This report is a boundary review and conceptual synthesis, not a claim of exhaustive bibliometric coverage. It combines the uploaded foundational corpus with selected primary and canonical literature chosen to test the proposed explanatory boundaries.

The uploaded corpus covers ecological perception, cognitive engineering, situated action, distributed cognition, activity theory, interaction design, everyday design, and computational modeling. It was assessed under the accompanying research canon: concepts must arise from necessity, remain implementation-independent, distinguish observation from inference, be measurable in principle, be falsifiable, and add to a shared ontology rather than duplicate vocabulary.

Two source limitations were handled explicitly. The uploaded file titled *Cognition in the Wild* is a review rather than the complete book, so claims about distributed cognition were checked against Hutchins's primary formulations and the later distributed-cognition program in HCI. The uploaded Engeström file is a critical essay about activity theory, so the review distinguishes Engeström's primary framework from that critique (Blunden, 2015).

The analysis proceeds in four stages:

6. identify what each adjacent field treats as its explanatory unit and mechanism;
7. identify where each field stops or becomes weak;
8. map the proposed candidate concepts onto established terminology;
9. retain only residual additions that support discriminating empirical tests.

4. Literature review and explanatory boundaries

4.1 Boundary matrix

Discipline or program	Primary contribution, assumptions, and strengths	Boundary, overlap, and residual gap
Human-Computer Interaction	Studies the design, use, and consequences of interactive computing. It contributes experimental methods, field methods, design research, usability and experience constructs, interaction techniques, and a large body of models. Its strength is direct engagement with computational artifacts and the diversity of settings in which they are used (Hornbæk & Oulasvirta, 2017; Hornbæk et al., 2019; Hornbæk, Kristensson, & Oulasvirta, 2025).	HCI already contains most of the proposed domain. Its weakness is not absence of coverage but theoretical pluralism: "interaction" is used as dialogue, transmission, tool use, embodiment, optimal behavior, and other concepts. HCI frequently mixes explanation, evaluation, and prescription. The residual gap is a shared, testable integration across paradigms and time scales, not a new subject matter.
Cognitive engineering and classical quantitative HCI	Card, Moran, and Newell developed the Model Human Processor, GOMS, the keystroke-level model, unit-task analysis, and explicit links between cognitive theory and engineering prediction. The approach assumes decomposable goals, operators, methods, selection rules, and bounded processing stages. It is strongest for practiced, routine, measurable tasks (Card et al., 1983; Newell & Card, 1985).	It is deliberately narrow: expert performance, temporal prediction, and tasks that can be decomposed reliably. It handles neither open-ended interpretation nor institutional and cultural organization well. Its "operators" are useful model-relative units, not universal interaction primitives.
Human factors and cognitive systems engineering	Explains performance and safety in work systems through workload, attention, control, automation, error, situation awareness, skill-rule-knowledge modes, and joint cognitive systems. It assumes that human and technical components should be analyzed as an operational system under constraints. It is strong for high-consequence, time-critical, and adaptive work (Rasmussen, 1983; Vicente, 1999; Parasuraman et al., 2000; Hollnagel & Woods, 2005).	It can understate meaning, identity, culture, and everyday appropriation when performance and control dominate the analysis. Cognitive work analysis and joint cognitive systems already cover many proposed context and system concepts. The residual problem is linking operational models to semantic and longitudinal learning accounts without reducing either.
Cognitive psychology	Provides experimentally grounded mechanisms for perception, attention, memory, decision, problem solving, skill acquisition, automaticity, mental models, and transfer. It assumes behavior can be explained through measurable cognitive processes and representations, although schools disagree about their form (Norman, 1983; Logan, 1988; Singley & Anderson, 1989).	Laboratory tasks and individual-level models can neglect distributed resources, social organization, historical practice, and institutional constraints. Still, accumulated experience, strategy change, learning curves, mental models, and transfer are established topics; they do not require new umbrella terms.
Cognitive science	Integrates psychology, neuroscience, linguistics, philosophy, artificial intelligence, and computational modeling. It supplies representational, embodied, Bayesian, dynamical, and resource-rational accounts of cognition. Its strength is mechanistic plurality and	It is broader than interaction and may not prioritize the special causal role of computational artifacts. However, its frameworks already express latent state, partial observability, learning, action selection, and representation. The residual gap is domain-specific

Discipline or program	Primary contribution, assumptions, and strengths	Boundary, overlap, and residual gap
	formal model comparison (Chemero, 2009; Gershman et al., 2015; Farrell & Lewandowsky, 2018).	integration and validation, not missing formal machinery.
Ecological psychology	Gibson treats perception and action as relations between an organism and a structured environment. Affordances are agent-environment relations, and perceptual information can specify action possibilities without requiring a reconstruction of the physical world. It is strong for embodiment, real-time coupling, and the relational nature of action possibilities (Gibson, 1979; Chemero, 2009).	Strong direct-perception claims remain contested, and the framework is less complete for conventional symbols, propositional meaning, hidden computational states, and institutional rules. It supplies a necessary relational correction but not a complete theory of computational interaction.
Situated action, ethnomethodology, and conversation analysis	Suchman shows that plans are resources for situated action rather than complete determinants of conduct. Ethnomethodology and conversation analysis explain how intelligibility, sequencing, common ground, trouble, and repair are locally produced. These approaches assume action is indexical and accountable within unfolding circumstances (Sacks et al., 1974; Clark & Brennan, 1991; Suchman, 1987).	They provide exceptional sensitivity to context and the underdetermination of intention by observable behavior. They are less oriented toward cross-domain quantitative prediction and formal learning models. Their major implication is methodological: event logs do not reveal meaning without the circumstances and practices that make actions intelligible.
Distributed cognition	Hutchins and later HCI work relocate cognition from an isolated individual to systems of people, artifacts, representations, and environments. Representational states propagate across media and time; external structures transform computational work. It is strong for teams, navigation, external memory, workflow, and artifact-mediated reasoning (Hutchins, 1995; Zhang & Norman, 1994; Hollan et al., 2000).	The analyst must choose system boundaries, and accounts may understate power, emotion, identity, or contested meaning. Distributed cognition already explains why interaction cannot be inferred from interface events alone. The residual task is to connect its system-level descriptions to measurable individual learning and transfer.
Activity theory	Activity theory explains object-oriented, culturally mediated activity through subjects, tools, community, rules, division of labor, and historically developing contradictions. It is strong for collective work, institutional context, motive, and developmental change (Engeström, 1987; Kuutti, 1996; Kaptelinin & Nardi, 2006).	Its unit of analysis is broad, and operationalization and quantitative prediction are difficult. Engeström's activity-system diagram is a useful heuristic but should not be treated as a complete social theory or a micro-model of interaction. It overlaps heavily with proposed behavioral context and adaptation constructs.
Situated cognition and embodied cognition	These programs argue that cognition depends on bodily action, environmental structure, and task context rather than only internal symbol manipulation. They explain offloading, sensorimotor regularities, and the role of material arrangements (Lave & Wenger, 1991; Dourish, 2001; Chemero, 2009).	They vary from precise dynamical models to broad philosophical positions. Their concepts can become too general unless tied to explicit observables. They reinforce the need for joint-state and environment variables but do not uniquely define a science of interaction.
Learning sciences and skill-transfer	Skill acquisition, instance learning,	Transfer is often difficult to predict

Discipline or program	Primary contribution, assumptions, and strengths	Boundary, overlap, and residual gap
research	proceduralization, transfer of training, situated learning, apprenticeship, and communities of practice explain how experience changes future performance. Transfer research connects source and target tasks through shared components, productions, principles, or participation structures (Logan, 1988; Singley & Anderson, 1989; Lave & Wenger, 1991).	because similarity is multidimensional and representation-dependent. This is a genuine empirical gap, but the constructs already exist. The needed addition is a typed similarity and equivalence schema plus stronger longitudinal designs.
Information visualization and graphical perception	Studies how visual encodings, spatial organization, perceptual tasks, and external representations affect accuracy, search, inference, and reasoning. It establishes that informationally equivalent displays may impose different computational costs and that elementary visual judgments differ in precision (Cleveland & McGill, 1984; Larkin & Simon, 1987).	Its domain is narrower than interaction as a whole and often assumes a defined analytic task. It already explains why visual structure can affect action before full task interpretation, but it does not support a simple visual-versus-semantic serial pipeline.
Information architecture and human information interaction	Explains organization, labeling, navigation, search, findability, information behavior, and the relation between information practices and systems. It assumes that people act within structured information environments and use multiple strategies to seek, avoid, assess, and organize information (Pirolli & Card, 1999; Fidel, 2012; Rosenfeld et al., 2015).	Information architecture is often prescriptive and practice-oriented rather than a general causal science. Human information interaction is broader and ecological but focuses on relations with information rather than all computational action. These fields cover much of semantic organization and wayfinding without requiring new terminology.
Semiotics and semiotic engineering	Distinguishes signs, objects, interpretants, codes, and meaning-making. Semiotic engineering treats an interface as a medium through which designers communicate a model of intended use to users. It is strong for the distinction between representation and meaning and for failures of interpretation (de Souza, 2005).	Semiotic analysis is often interpretive and may lack discriminating quantitative predictions. Meaning is not exhausted by designer communication, especially in adaptive, appropriated, or multi-agent systems. It nevertheless makes a new "semantic representation" construct unnecessary unless the proposed term is more precise.
Systems theory, cybernetics, and control	Supplies feedback, state, transition, regulation, stability, adaptation, observability, and requisite variety. It treats behavior as trajectories of coupled systems rather than isolated events. This is the clearest existing formal basis for reciprocal interaction (Wiener, 1948; Ashby, 1956; von Bertalanffy, 1968).	Generic systems language can become vacuous unless variables and mechanisms are domain-specific. Classical control is often thin on interpretation, social meaning, and learning. The residual challenge is not feedback itself but integration with cognition, representation, and culture.
Dynamical systems and complexity science	Explains temporally extended, nonlinear, emergent, path-dependent, and multi-scale behavior. It supports analysis of attractors, coordination, phase transitions, and development without assuming fixed symbolic programs (Simon, 1962; Thelen & Smith, 1994).	Complexity vocabulary is easy to use metaphorically and does not identify the correct state variables by itself. It contributes methods and cautions, but no distinctive interaction ontology follows merely from calling interaction complex.
Anthropology and ethnography	Provides thick description of practice, culture, meaning, materiality, identity,	Generalization is analytic rather than usually law-like, and comparative

Discipline or program	Primary contribution, assumptions, and strengths	Boundary, overlap, and residual gap
	institutions, and local categories. Ethnographic work in HCI reveals how technologies are appropriated and how formal system descriptions diverge from actual practice (Geertz, 1973; Suchman, 1987; Dourish, 2001).	measurement can be difficult. Anthropology already supplies the antidote to demographic determinism and context-free interpretation. The residual task is principled translation between local explanations and cross-setting models.
Behavioral economics and decision science	Models bounded choice, reference dependence, heuristics, framing, uncertainty, learning, incentives, and choice architecture. Its central strength is explicit comparison between available alternatives, beliefs, utilities, and observed choices (Simon, 1955; Kahneman & Tversky, 1979).	Choice models cover only part of interaction and can ignore perception-action dynamics, social practice, and meaning. Still, they make "behavior relative to available actions" an established idea. Any interaction model should distinguish objective, signaled, and believed choice sets.
Interaction design and design research	Produces artifacts, methods, experiential accounts, and prescriptive knowledge about how interactions might be configured. It is strong at generating possibilities, discovering requirements, and exposing phenomena through making (Rogers et al., 2011; Norman, 2013).	Design knowledge is not automatically explanatory. A successful artifact demonstrates possibility, not a general mechanism. Interaction design should remain a major source of hypotheses and interventions, while descriptive theory and prescription remain explicitly separate.
Computational interaction and computational cognitive modeling	Uses formal models, optimization, simulation, probabilistic inference, reinforcement learning, and model comparison to explain behavior and improve interactive systems. Resource-rational and bounded-agent approaches connect environmental structure, human limitations, and adaptive policy (Oulasvirta et al., 2018; Oulasvirta, Jokinen, & Howes, 2022).	Existing models often concentrate on tasks with tractable objectives and measurable performance, and they remain weaker on social institutions, contested meaning, identity, and historical practice. This program is the closest existing vehicle for the proposed formal science; creating a separate discipline before extending it would be premature.

4.2 What the literature collectively explains

Taken together, the adjacent fields already explain the main phenomena in the working direction:

- **Experience affects later behavior** through learning, memory, mental models, procedural skill, automaticity, habit, and transfer.
- **Current events underdetermine meaning** because observable action admits multiple intentions and because cognition may be distributed across people, artifacts, and circumstances.
- **Behavior is conditional on alternatives and constraints** through affordances, choice sets, bounded rationality, task analysis, and control.
- **Representation and meaning are distinguishable** through semiotics, distributed representations, graphical perception, and external cognition.
- **Visual structure changes behavior** through perceptual organization, search, encoding accuracy, spatial indexing, and computational cost.
- **Interaction has formal structure** through feedback loops, state-transition systems, GOMS operators, control models, conversation sequences, and activity hierarchies.

- **Context is constitutive rather than decorative** through situated action, anthropology, activity theory, ecological psychology, and distributed cognition.
- **Social organization changes cognition and action** through common ground, division of labor, rules, roles, and institutional objects.

No candidate phenomenon is presently an unexplained residue shared by no existing field.

4.3 Where the literature collectively stops

The literature is weaker at four integrations.

4.3.1 Cross-timescale integration

Momentary perception and action, episode-level strategy, long-term learning, and historical or institutional change are usually modeled separately. A theory needs explicit mappings between them. Prior history should influence present action through identifiable current states - beliefs, skill, expectations, habits, social position, or access - rather than through an unanalyzed history variable.

4.3.2 Typed representation equivalence

Research shows that the same information can be represented in behaviorally different ways, but studies often use "equivalent" without specifying the relation. Semantic equivalence, informational equivalence, task equivalence, computational equivalence, and interactional equivalence are not interchangeable.

4.3.3 Context-conditioned behavioral evidence

Event traces are routinely analyzed without a sufficient record of what the person could perceive, what actions were feasible, what actions were signaled, what the person believed possible, or what social constraints applied. This makes intention and strategy weakly identifiable (Dey, 2001; Suchman, 1987).

4.3.4 Cross-scale predictive comparison

Contextual and social theories often explain cases richly but do not compete in shared predictive benchmarks. Formal models often predict constrained tasks but omit social and historical variables. There is no standard benchmark that tests whether adding those variables improves held-out prediction across implementations.

These are integration and evaluation gaps. They do not yet establish a new discipline.

5. Critique of the current working direction

5.1 "Interaction behavior develops through accumulated experience interacting with information systems"

Disposition: established but underspecified. Skill acquisition, instance learning, procedural knowledge, mental models, transfer, and situated learning already cover this claim (Norman, 1983; Logan, 1988; Singley & Anderson, 1989; Lave & Wenger, 1991). "Accumulated interaction experience" is too broad to function as a mechanism. Two people can have equal exposure counts but different learned states because the experiences differed in feedback, goals, attention, success, spacing, social support, and interpretation.

The stronger formulation is:

> Prior interaction episodes affect future behavior only insofar as they change currently operative states or environments, such as beliefs, skills, expectations, habits, perceived action possibilities, social roles, or access to external resources.

This formulation is falsifiable. If history has no residual effect after those mediators are measured adequately, a separate history construct is unnecessary. If a residual persists, the scientific task is to identify the missing mediator rather than preserve history as an opaque cause.

5.2 "Previous interaction experiences influence future interaction"

Disposition: true but nearly tautological without conditions. The proposition needs a defined source experience, target interaction, causal contrast, mechanism, direction, and time interval (Singley & Anderson, 1989). Experience can improve, impair, or leave target performance unchanged. Familiar cues can produce negative transfer when mappings reverse. Forgetting, interference, and contextual specificity constrain persistence.

Use the established term **transfer of learning** or **transfer of skill**. Define it counterfactually as the change in target behavior caused by prior source experience under otherwise comparable target conditions.

5.3 "Interaction cannot be fully understood from interface events alone"

Disposition: generally correct, but "fully" is untestable and the boundary is task-dependent. Situated action, distributed cognition, ethnomethodology, and cognitive modeling all show that the same observable event can arise from different intentions, strategies, beliefs, or circumstances (Card et al., 1983; Suchman, 1987; Hollan et al., 2000). However, event-only models can be adequate for narrow prediction problems such as practiced pointing or transcription.

The defensible claim is:

> An event trace is insufficient for any inference whose competing mechanisms predict the same event distribution under the recorded variables.

This turns a slogan into an identifiability criterion. More context is required only when it discriminates among plausible explanations or supports the target intervention.

5.4 "Behavior should be interpreted relative to the information, actions, and constraints available at the time"

Disposition: strongly supported, with one necessary refinement. Affordance, signifier, bounded-choice, and control traditions all require action to be interpreted against opportunities and constraints (Simon, 1955; Gibson, 1979; Norman, 2013). "Available" must be split into at least three sets:

- **feasible action set:** actions physically and computationally permitted;
- **signaled action set:** actions made perceptually discoverable by the current representation;
- **believed action set:** actions the person represents as possible.

These sets can differ. A function may be feasible but not signaled; signaled but not understood; believed possible but unavailable; or socially prohibited despite technical feasibility. Behavior should be interpreted against the relevant set for the proposed mechanism.

5.5 "Semantic understanding may exist independently of visual representation"

Disposition: reject the strong independence claim; retain the distinction. Meaning and visual form are analytically distinct, and the same content can be expressed through visual, auditory, tactile, linguistic, or procedural representations (Zhang & Norman, 1994; de Souza, 2005). But human semantic understanding is always mediated by some perceptible or remembered representation and by prior knowledge. It is therefore misleading to say that meaning exists independently of representation in the psychological process.

Use three terms:

- **representation:** a perceptible or manipulable form that can carry distinctions;
- **content:** what a representation is taken to be about under a code or practice;
- **interpretation:** an agent's situated mapping from representation to believed content, implication, or action.

This separation preserves the intended insight without treating meaning as a free-standing object.

5.6 "Visual structure may influence interaction before semantic understanding is fully formed"

Disposition: plausible and partly supported, but not as a universal serial stage. Graphical perception and attention research show that some visual features and spatial relations influence detection, search, and action rapidly (Treisman & Gelade, 1980; Cleveland & McGill, 1984; Larkin & Simon, 1987). Larkin and Simon show that informationally equivalent diagrams can reduce search and inference costs. Yet semantic expectations can also guide early attention, and expertise changes what structure is perceived.

The testable version is:

> Under specified exposure, task, and expertise conditions, differences in visual encoding produce early differences in attention or action that remain after semantic content is held constant.

This is an empirical hypothesis about timing and mediation, not a foundational axiom.

5.7 "Interaction may be describable through a stable set of primitives independent of specific software"

Disposition: unproven and likely false if "primitive" means universal cognitive atom. Existing traditions propose different units because they answer different questions (Sacks et al., 1974; Card et al., 1983; Kuutti, 1996): keystrokes and mental operators in GOMS, perception-action cycles in ecological psychology, turns and repair sequences in conversation analysis, actions and operations in activity theory, and state transitions in control models.

No context-free smallest meaningful unit has been established. The minimum defensible instrumentation unit is a **closed-loop transition record**: prior state and opportunities, one or more agent actions, resulting transition, and observable consequence. This is a data schema, not a claim that cognition is composed of such atoms.

6. Recommended terminology

6.1 Candidate-term disposition

Candidate term	Recommendation	Reason
Accumulated interaction experience	Replace with **interaction history** for records and **learned state** for mechanisms	Exposure is not a mechanism; current beliefs, skills, expectations, habits, and social position mediate effects.
Visual structure	Use **visual encoding**, **perceptual organization**, or **spatial layout** as appropriate	Established terms are more specific and measurable.
Interaction primitives	Use **model-relative operators** or **state-transition roles**; reject a universal primitive claim	The correct grain depends on the explanandum.
Semantic representation	Split into **representation**, **content**, and **interpretation**	Prevents form, meaning, and agent understanding from being conflated.
Behavioral context	Use **task, environmental, social, and institutional context**, explicitly enumerated	"Context" is otherwise an unbounded residual category.
Interaction strategies	Use **strategy**, **policy**, or **method** and state which theory is intended	These terms have distinct formal meanings.
Adaptation and learning	Retain established terms; distinguish within-episode adaptation, learning, development, and system adaptation	Different time scales and mechanisms should not be collapsed.
Behavioral evidence	Use **behavioral trace** for observation and **evidence under an observation model** for inference	Data are not evidence for a claim until linked through a validity argument.
Interaction measurement	Retain as a methodological area, not a construct	Measurement concerns many constructs and does not define a new object.
Interaction modeling	Retain as a research activity; specify cognitive, control, statistical, social, or computational model	The phrase alone states no theory.
Representation equivalence	Replace with **typed equivalence relation**	Equivalence must specify semantic, informational, task, computational, or interactional criteria.
Interaction similarity	Use a **similarity profile** over declared dimensions, not a single scalar by default	Surface, semantic, operational, temporal, and social similarity can diverge.
Interaction transfer	Use **transfer of learning** or **transfer of skill in human-computational interaction**	The phenomenon is established; the domain qualifier is sufficient.

6.2 Terms that should not be introduced

Do not coin new terms for mental models, affordances, feedback, constraints, strategy, transfer, skill, external representation, common ground, repair, context, or distributed cognition unless an empirical result demonstrates that the established construct cannot express the required distinction.

7. Canonical ontology

The ontology below is intentionally small. It is not a claim that every theory must use every entity. It is a shared translation layer that permits theories to state which entities they include, omit, or redefine.

7.1 Entities and relations

Concept	Canonical definition	Typical operationalization	Status
Human agent	A person or coordinated human group whose actions and learned states participate in the interaction trajectory.	Participant identity, role, capabilities, group membership, coordination structure.	Established.
Computational process	A rule-governed, stateful process that transforms information or acts in an environment and can condition later human observations or opportunities.	State machine, probabilistic policy, service, algorithm, controller, model, or distributed process.	Established.
Environment	Physical, informational, social, and institutional conditions not assigned to the focal agents but causally relevant to the trajectory.	Device setting, location, surrounding artifacts, norms, organization, other actors, environmental disturbances.	Established.
Joint state	The external state of the coupled human-computational-environmental system at a specified time and grain.	Synchronized system state, artifact state, task state, location, active roles, resource availability.	Synthesis of systems concepts.
Goal	A represented desired condition that guides action selection.	Instruction, stated goal, inferred subgoal, reward structure.	Established.
Task	A normative or analytic specification of conditions, transformations, and completion criteria.	Task protocol, work demand, benchmark.	Established.
Object or motive	The socially or historically meaningful object toward which collective activity is organized.	Work objective, institutional purpose, community practice.	Established in activity theory; not reducible to an immediate goal.
Representation	A perceptible or manipulable form that preserves distinctions and can support interpretation or action.	Text, sound, spatial arrangement, haptic pattern, physical artifact, state display, conversational utterance.	Established.
Content	What a representation is taken to concern under a code, convention, or practice.	Proposition, referent, data relation, domain state.	Established.
Interpretation	An agent-relative mapping from representation and context to believed content, implication, or action relevance.	Comprehension response, prediction, explanation, categorization, action choice, model-based latent variable.	Established synthesis.
Observation	Information available to an agent about joint state through current channels.	Rendered display, sound, haptic feedback, system message, remembered or externally stored cue.	Established.
Feasible action set	Actions permitted by current physical, computational, and formal constraints.	Enumerated commands, enabled controls, reachable states, API permissions.	Established choice/control concept.

Concept	Canonical definition	Typical operationalization	Status
Signaled action set	Feasible or infeasible actions made perceptually discoverable by the representation.	Visible controls, prompts, signifiers, menu entries, affordance cues.	Synthesis of signifier and action-set concepts.
Believed action set	Actions the human agent currently represents as possible or relevant.	Elicitation, prediction from behavior, recognition test, mental-model probe.	Established belief/choice concept.
Constraint	A relation that removes, penalizes, delays, or socially prohibits possible states or actions.	Physical lockout, validation rule, time pressure, policy, norm, cost, capability limit.	Established.
Action	An intervention by an agent that can change joint state or the information available to another agent.	Motor action, command, utterance, delegation, system-initiated operation.	Established.
Transition	A change in joint state generated by prior state, agent actions, and disturbances.	Logged state change, computational transition, physical consequence.	Established.
Feedback	An observation whose value depends causally on a prior action or state and can regulate subsequent action.	Confirmation, error, movement, latency signal, consequence, social response.	Established cybernetic concept.
Outcome	A state or measure used to evaluate an episode relative to a task, goal, value, or analytic question.	Completion, error, time, utility, safety, learning, coordination quality.	Established.
Strategy or policy	A mapping, explicit or implicit, from represented state and goals to action selection over time.	Sequence model, verbal protocol, policy parameters, GOMS method, inferred latent strategy.	Established.
Learned state	Relatively persistent agent state changed by experience and capable of affecting later interpretation or action.	Beliefs, mental model, procedural knowledge, skill parameters, habits, expectations, learned policy.	Established umbrella for model integration.
Interaction history	A temporally ordered record of prior episodes and consequences, whether or not retained by the agent.	Longitudinal trace, exposure record, training history.	Descriptive record, not mechanism.
Episode	An analytically bounded segment of a trajectory selected by a declared criterion.	Task trial, conversational sequence, work shift, session, project phase.	Necessary analytic convention; not a natural atom.
Breakdown and repair	A failure of expected coordination or intelligibility and the actions through which participants detect, diagnose, or restore it.	Error recovery, clarification sequence, undo, help-seeking, handoff repair.	Established.
Transfer	A causal change in target performance or strategy attributable to prior source experience.	Matched-control difference, longitudinal crossover, retention and generalization tests.	Established.
Behavioral trace	A time-ordered record of observable actions and states produced during an episode.	Event log, video, motion trace, conversational transcript.	Observation.

Concept	Canonical definition	Typical operationalization	Status
Observation model	An explicit account of how latent constructs and processes could generate the recorded data, including omissions and error.	Measurement model, coding scheme, sensor model, likelihood, qualitative warrant.	Established scientific requirement.
Evidence	Observations that discriminate among claims under an explicit observation model and validity argument.	Model comparison, triangulated case evidence, intervention effect, replication.	Epistemic relation, not raw data.

7.2 Typed equivalence relations

A single relation called "representation equivalence" is too coarse. Use the following types.

Type	Definition	Example consequence
Syntactic equivalence	Same tokens or structural encoding under a declared grammar.	Useful for implementation comparison; says little about meaning or use.
Informational equivalence	The same task-relevant information can in principle be recovered.	Two displays may be informationally equivalent yet impose different search and inference costs.
Semantic equivalence	The representations express the same referent, proposition, or domain relation under a declared interpretive scheme.	Does not imply equal discoverability or performance.
Task equivalence	The same goal and success conditions apply.	Does not imply the same action sequence or cognitive demands.
Computational equivalence	The same transformations can be completed with comparable resource requirements under a declared model.	Stronger than informational equivalence and model-dependent.
Interactional equivalence	Two conditions yield equivalent distributions of strategies, actions, outcomes, or learning for a specified population and context within a tolerance.	Empirical and explicitly population-, context-, and metric-bound.

Equivalence in one type does not entail equivalence in another. The distinction directly explains why semantically identical controls can behave differently, why visually different systems can support transfer, and why surface similarity can create negative transfer.

7.3 Interaction similarity profile

Interaction similarity should normally be represented as a vector rather than a single score. Candidate dimensions are:

- goal and task structure;
- semantic relations;
- state-transition topology;
- action-effect mappings;
- required perceptual discriminations;
- required cognitive operations or productions;
- feedback timing and reliability;

- physical or motor demands;
- error and repair structure;
- social roles, norms, and division of labor;
- representation conventions;
- reward, risk, and institutional consequences.

A scalar similarity measure is justified only after a model shows how dimensions combine for the target prediction.

8. Foundational assumptions

These assumptions define the proposed integration program. They are not asserted as universal truths beyond its scope.

10. **Reciprocal coupling:** interaction requires a causal loop, although influence can be delayed, asymmetric, or mediated.
11. **Partial observability:** neither the human nor the computational process necessarily has complete access to joint state, intentions, or consequences.
12. **Bounded agency:** action is constrained by perception, cognition, embodiment, time, resources, permissions, and social organization.
13. **Historical mediation:** prior episodes matter through changed learned state, external resources, relationships, or institutions.
14. **Representational mediation:** computational state affects human action through representations and channels, and human action affects computation through encodings and sensors.
15. **Interpretive variability:** the same representation can support different interpretations across people, times, and practices.
16. **Multiple time scales:** perception-action coupling, strategy, learning, development, organizational change, and cultural change may obey different dynamics.
17. **Context specificity:** explanatory relevance is claim-dependent; context must be enumerated rather than treated as an unlimited background.
18. **Social causation:** norms, roles, language, power, access, and institutions can be causal components, not merely demographic descriptors.
19. **Implementation independence:** core claims refer to relations and mechanisms, not named technologies or interface conventions.
20. **Measurement dependence:** latent constructs are not directly observed; every inference depends on an observation model and validity evidence.
21. **Heterogeneity:** population averages do not automatically describe individual mechanisms or subgroup distributions.

9. First principles

22. **Interaction is a trajectory, not an isolated event.** Events acquire explanatory meaning through prior state, alternatives, consequences, and subsequent adaptation.
23. **Behavior must be conditioned on opportunities.** An action cannot be interpreted without distinguishing what was feasible, signaled, believed possible, and socially permitted.
24. **History is not a mechanism.** Prior experience should be decomposed into the states and external structures through which it affects present behavior.
25. **Representation is not meaning.** Form, content, and situated interpretation are distinct, even though human meaning is always representationally mediated.

26. **Equivalence is typed.** No claim of sameness is complete without a declared relation, population, context, metric, and tolerance.
27. **Primitives are model-relative.** The smallest useful unit is determined by the explanatory question; no universal cognitive atom should be presumed.
28. **Context is bounded by contrast.** Include a contextual variable when it changes a prediction, intervention, or interpretation among plausible alternatives.
29. **Social categories are not default mechanisms.** Demographic variables require causal pathways, measurement invariance, and attention to within-group heterogeneity.
30. **Explanation precedes prescription.** A design recommendation is not evidence that the proposed mechanism is correct.
31. **New constructs require incremental success.** A term earns canonical status only when it explains or predicts something that established constructs cannot with equal parsimony.

10. Relationships among concepts

10.1 Core causal cycle

At a declared time scale, goals, learned state, and context shape an agent's interpretation of observations. Interpretation and believed action possibilities shape strategy and action. Human and computational actions jointly produce a state transition. The new state is represented through perceptual channels as feedback. Feedback and outcome update learned state and may modify the external environment, social relations, or computational policy. Repeated transitions form an episode; repeated episodes create an interaction history.

10.2 Dependency graph

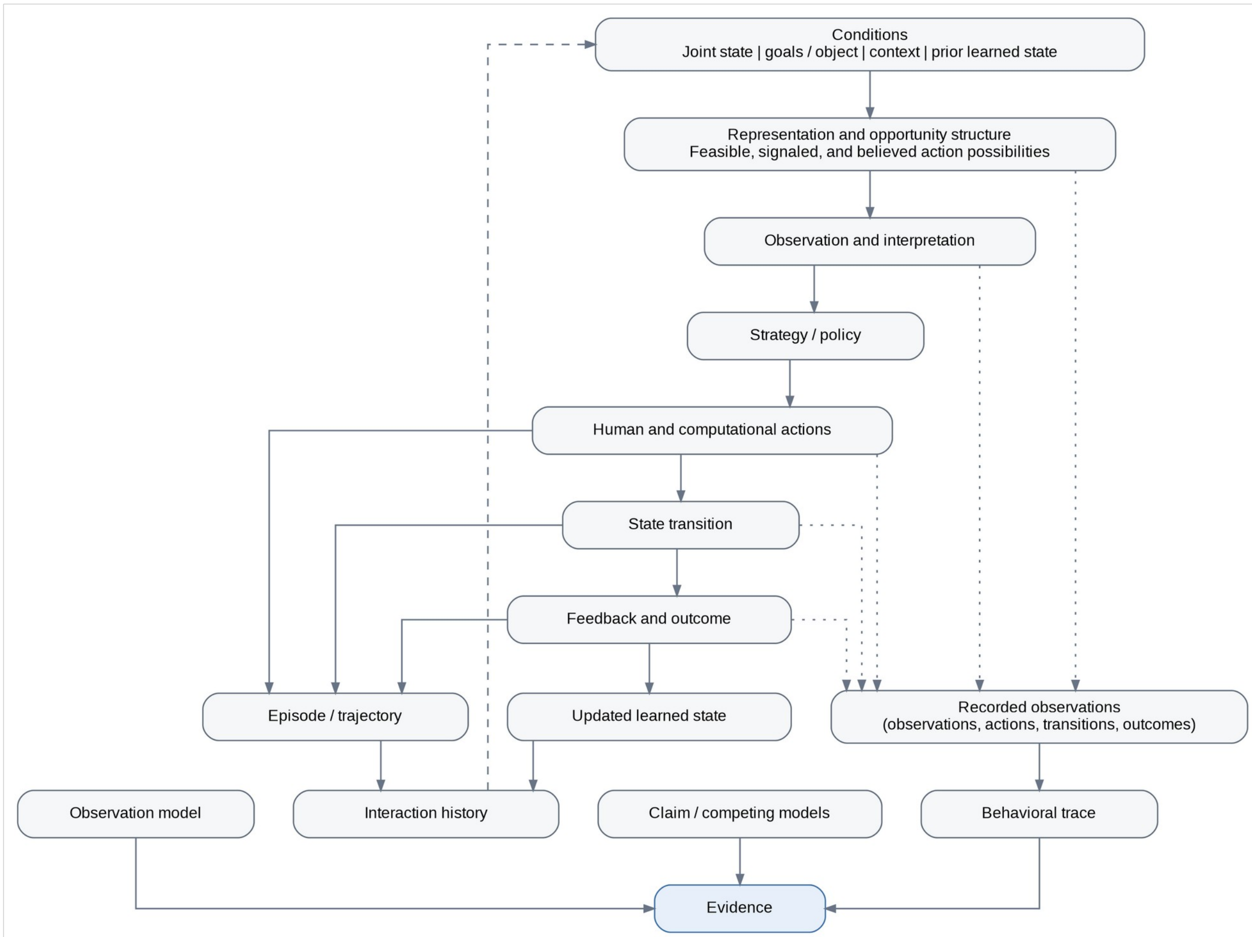

Figure 1. Dependency graph for the minimum ontology. Arrows indicate explanatory dependence, not necessarily temporal precedence.

The graph has two important asymmetries. First, interaction history is upstream of current behavior only through learned state or changed external conditions. Second, behavioral traces become evidence only through an observation model and an explicit claim; a trace is not self-interpreting.

11. Minimum formal model

A useful integration model should be general enough to compare theories but constrained enough to generate discriminating predictions. The following is a template, not a claim that humans literally implement a particular control or reinforcement-learning algorithm.

Let:

- x_t be joint external state at time t;
- z_t be the human learned state, including beliefs, skill, expectations, and strategy parameters;
- q_t be task, social, and institutional context;
- r_t be the representation generated from current state;
- F_t, S_t, and B_t be feasible, signaled, and believed human action sets;
- h_t be human action;
- c_t be computational action, including autonomous or delayed action;

- `o_t` be the human observation produced through representation and environment;
- `y_t` be an outcome measure.

A minimal coupled process is:

`r_t = R(x_t, q_t)`

`o_t ~ O_H(r_t, x_t, q_t)`

`B_t ~ I(o_t, z_t, q_t)`

`h_t ~ pi_H(h | o_t, B_t, z_t, q_t, goal_t)`

`c_t ~ pi_C(c | x_t, h_<=t, q_t)`

`x_(t+1) ~ T(x_t, h_t, c_t, q_t, disturbance_t)`

`y_t = Y(x_t, h_t, c_t, x_(t+1), q_t)`

`z_(t+1) ~ U(z_t, o_t, h_t, c_t, y_t, q_t)`

This template makes several empirical questions explicit:

- Does the representation change behavior through observation, interpretation, or both?
- Does prior history add predictive value after `z_t` is measured?
- Do social or institutional variables change the policy, transition, or outcome function?
- Are two representations equivalent under `R`, `I`, the policy, or only the outcome metric?
- Does a theory predict held-out trajectories when the implementation changes but the declared structure remains constant?

11.1 Why this is not a new theory by itself

Every component has precedents in control theory, partially observed decision processes, cognitive modeling, ecological action, distributed cognition, and HCI. Its contribution is a common interface among theories. Scientific value arises only if specific submodels generate risky predictions and outperform alternatives.

12. Smallest meaningful unit of interaction

There is no theory-independent smallest meaningful unit.

- For keystroke-level timing, an operator can be useful.
- For conversational intelligibility, a turn, adjacency pair, or repair sequence may be minimal.
- For ecological control, a perception-action cycle or continuously coupled variable is more appropriate.
- For activity theory, an operation has meaning only within an action, and an action only within an activity.
- For distributed cognition, the relevant unit may span people and artifacts.

The recommended minimum **recording unit** is a closed-loop transition:

`<pre-state, observations, action sets, agent action(s), transition, feedback, outcome, provenance>`

It is the smallest record that can support a causal interaction claim without treating an input event as self-explanatory. It is not asserted to be the smallest psychological or social unit.

13. What constitutes evidence in interaction research

13.1 Observation is not evidence by itself

A click, utterance, error, gaze fixation, or completion time is an observation. It becomes evidence for a claim only when:

- the construct and competing explanations are specified;
- the observation process and missing data are understood;
- the relevant alternatives and constraints are recorded or experimentally controlled;
- the measure has a validity argument;
- the analysis discriminates among models or causal contrasts;
- uncertainty and heterogeneity are reported.

13.2 Context-complete interaction record

A reusable empirical record should contain, when relevant:

Component	Minimum content
Provenance	Clock, episode, participant or role, device/channel, data version, synchronization uncertainty.
Pre-state	Computational state, task state, active representation, environmental and social state.
Opportunity structure	Feasible, signaled, and, when measured, believed action sets; permissions and constraints.
Action	Actor, action type, parameters, onset, duration, confidence or coding uncertainty.
Transition	Resulting state change, latency, system initiative, stochasticity, external disturbances.
Feedback	What became perceptually available to each agent and when.
Outcome	Task, safety, utility, learning, coordination, or experiential measure with declared validity.
History	Relevant prior exposures, training, expertise, recent episodes, and external resources.
Context	Goal, motive, language, role, norms, organization, location, stakes, collaboration.
Interpretation evidence	Comprehension, prediction, explanation, recall, self-report, or qualitative account, kept distinct from behavior.

Not every study needs every field. The rule is contrastive: retain the variables required to distinguish the mechanisms under test.

13.3 Evidence hierarchy by claim type

There is no universal hierarchy in which laboratory experiments always outrank field evidence. Appropriate evidence depends on the claim.

- **Causal effect of a representation:** randomized or strongly identified intervention, with manipulation checks and process measures.
- **Mechanism of skilled action:** quantitative model comparison, perturbation, and out-of-sample prediction.
- **Meaning and local intelligibility:** interaction analysis, ethnography, conversation analysis, and participant accounts tied to recorded conduct.
- **Longitudinal learning and transfer:** repeated measures, retention, matched controls, cross-system target tasks, and mediation analysis.
- **Collective cognition:** synchronized traces across people and artifacts, representational-state analysis, and field validation.
- **Institutional or cultural effects:** comparative and multilevel designs, measurement invariance, process evidence, and explicit treatment of power and access.

Triangulation is valuable when methods test different links in the same causal or interpretive argument, not merely when many data types are collected.

14. Universals, acquisitions, and population variation

14.1 Plausible near-universal constraints

The strongest candidates for broad generality are constraints rather than interface conventions or primitives:

- finite sensory resolution, attention, memory, motor precision, and time;
- dependence of control on sufficiently timely and informative feedback;
- sensitivity to environmental regularities and consequences;
- bodily and environmental constraints on possible action;
- partial observability and uncertainty;
- learning and forgetting over repeated experience;
- need for coordination and repair when action is jointly organized.

Even these vary quantitatively and can be altered by disability, development, tools, training, and environment. They should not be treated as fixed constants.

14.2 Primarily acquired components

- symbolic meanings and labels;
- reading direction and visual conventions;
- action-effect mappings;
- domain knowledge and mental models;
- skilled procedures and automatic routines;
- expectations imported from prior systems;
- social norms, roles, and authority;
- language-specific categories;
- occupational and institutional practices;
- strategies for search, verification, recovery, and collaboration.

Affordance perception often combines bodily constraints with learning. A possible action may be physically available but only discovered through cultural convention or prior experience.

14.3 Demographic variables

Demographic categories require measurement-invariance checks and causal decomposition rather than essentialist interpretation (VanderWeele & Robinson, 2014; Sen & Wasow, 2016; Millsap, 2011). They should be handled as follows:

32. **Describe sampling and inequity.** Age, disability, gender, race, ethnicity, socioeconomic position, and geography may be necessary to assess who is represented and who bears costs.
33. **Do not use categories as unexplained causes.** Replace deterministic statements with mechanisms such as access, discrimination, language proficiency, literacy, education, occupational practice, exposure, motor or perceptual capability, institutional role, and trust.
34. **Recognize social causation.** Categories can have causal consequences through institutions, discrimination, segregation, norms, and resource allocation. Calling them proxies must not erase those mechanisms.
35. **Test measurement invariance.** A metric or task may not measure the same construct across languages, cultures, ages, or accessibility conditions.
36. **Model heterogeneity.** Use multilevel or distributional models, interactions specified in advance, and within-group variation rather than only group means.
37. **Avoid post hoc essentialism.** Group differences do not establish inherent traits, and absence of an average difference does not establish equal experience or equal risk.
38. **Include affected communities in construct validation.** Especially when meaning, harm, or institutional practice is part of the claim.

Culture, language, education, occupation, and environment should enter models through observable practices and mechanisms where possible. They should not be reduced to fixed traits of people.

15. Learning, adaptation, and transfer

15.1 Required distinctions

- **Within-episode adaptation:** temporary adjustment to immediate feedback or state.
- **Learning:** a retained change attributable to experience.
- **Skill acquisition:** increasing reliability or efficiency in organized performance.
- **Automaticity or habit:** reduced dependence on deliberative control under recurring conditions.
- **Development:** longer-term reorganization of capabilities or participation.
- **Transfer:** an effect of source experience on a different target condition.
- **System adaptation:** change in the computational policy or representation, which can alter the human learning environment.

These processes can interact but should not be collapsed.

15.2 Transfer definition

For an outcome where higher values are better, define causal transfer as:

```
Transfer = E[Y_target | source experience] - E[Y_target | matched no-source condition]
```

For time or error, reverse or standardize the direction so positive values consistently indicate improvement. The estimate must specify retention interval, target representation, target task, population, and prior knowledge.

15.3 Mechanistic predictions

- Positive transfer should increase when source and target share required operations, causal structure, action-effect mappings, or generative principles.
- Surface similarity without shared structure should produce weak transfer.
- Familiar cues paired with changed consequences should produce negative transfer, especially under time pressure or automaticity.
- Explicit conceptual knowledge may transfer across representations even when motor procedures do not.
- Transfer should be moderated by expertise because experts encode deeper structure and novices may rely more on surface features.
- System adaptation can either scaffold learning or prevent stable learning if mappings and feedback change faster than the person can form reliable expectations.

These predictions are testable within existing learning and cognitive frameworks. An interaction-specific construct is warranted only if it explains residual transfer beyond them.

16. Representation, meaning, and visual structure

16.1 Representation-content-interpretation triad

A representation has physical or formal properties. Content is assigned under a code, convention, or practice. Interpretation is performed by an agent with a history and context. This triad prevents four common errors:

- treating an icon or layout as if it contained its meaning intrinsically;
- treating semantic equivalence as equal usability;
- inferring interpretation directly from action without testing alternatives;
- assuming that a visual redesign changes only appearance rather than computational work.

16.2 Visual structure

Visual structure can influence interaction through at least four mechanisms:

39. **Perceptual discriminability:** some encodings support more accurate elementary judgments.
40. **Attentional guidance:** salience and spatial organization change search order and detection.
41. **Spatial indexing:** diagrams can make relevant relations co-located and reduce search.
42. **External computation:** a representation can perform part of the organizational work that would otherwise require internal transformation or memory.

These effects can occur without complete semantic interpretation, but they remain task- and expertise-dependent. The appropriate experiments hold content constant, manipulate encoding, measure early process and later understanding separately, and test whether early effects mediate action.

17. Falsifiability and decisive tests

17.1 Claim-level falsification tests

Claim	Evidence that would count against it	Consequence
Prior interaction history has a distinctive effect	Once current beliefs, skill, expectations, recent exposure, and external resources are measured, history adds no held-out predictive value.	Retain current-state constructs; remove history as an explanatory variable.
Context-rich traces are necessary	Event-only models match context-rich models in out-of-domain prediction and intervention selection for the declared claim.	Context fields are unnecessary for that claim and grain.
Typed equivalence predicts behavior	Equivalence type and similarity dimensions fail to predict performance or transfer beyond surface features and task labels.	Revise or reject the taxonomy.
Visual structure has an early independent effect	Under controlled exposure and matched content, visual encoding has no reproducible effect on early attention or action after semantic knowledge is modeled.	Reject the claimed temporal mechanism.
A stable primitive set exists	The proposed primitives fail compositionality, cannot encode multiple modalities without ad hoc additions, or do not support cross-system prediction.	Treat units as model-relative.
The integration model is useful	A simpler established model predicts held-out trajectories and intervention effects equally well or better with fewer assumptions.	Reject the integration layer under parsimony.

17.2 Evidence that a new discipline is unnecessary

A new discipline should be rejected if a preregistered research program finds that:

- established constructs can be translated into the canonical ontology without explanatory loss;
- a composite model built from existing theories predicts interaction and transfer across domains;
- proposed new constructs add negligible, unstable, or non-replicating predictive value;
- no distinctive methods are required beyond existing experimental, computational, field, and interpretive methods;
- the most successful explanations remain domain-specific and do not converge on independent interaction laws;
- organizing the work within HCI and adjacent fields produces cumulative results without conceptual obstruction.

17.3 Evidence that could justify a new discipline

A positive case would require all of the following, not merely one:

43. a recurrent anomaly across multiple established theories and domains;
44. a new mechanism with precise scope and non-redundant predictions;

45. cross-implementation and cross-cultural replication;
46. superior held-out prediction and intervention selection against strong composite baselines;
47. measurement instruments and data standards that remain stable across technologies;
48. a progressive research program in which later results increase precision rather than multiply exceptions.

17.4 Recommended decisive benchmark

Construct a multi-laboratory benchmark with at least:

- three interaction modalities;
- three task families;
- multiple representations per semantic task;
- novice, intermediate, and expert participants;
- at least two linguistic or cultural settings;
- immediate, retained, and transfer outcomes;
- both individual and collaborative conditions;
- systems with different feedback latency, initiative, and predictability.

Compare four model classes on held-out systems and populations:

- M0: event-only statistical baseline;
- M1: strongest domain-specific established model;
- M2: integrated existing-theory model using the canonical ontology;
- M3: M2 plus any proposed interaction-specific construct.

Use calibration, predictive accuracy, intervention success, complexity penalties, parameter recoverability, and qualitative error analysis. The independent-discipline claim receives support only if M3 wins reproducibly and the added construct has a stable interpretation across domains.

18. Recommended sequence for foundational papers

Each paper should make one primary contribution and leave a reusable artifact.

Paper 1 - The discipline-necessity argument

Single contribution: a formal decision standard for whether human-computational interaction requires an independent discipline.

Contents: scope, boundary criteria, map of existing explanations, explicit negative current conclusion, and conditions for reversal.

Artifact: discipline test checklist and evidence matrix.

Paper 2 - Canonical ontology and terminology

Single contribution: a minimal shared ontology that maps established theories without claiming a new mechanism.

Contents: definitions, exclusions, typed equivalence relations, model-relative units, and mappings to HCI, cognitive science, human factors, distributed cognition, and activity theory.

Artifact: machine-readable ontology and crosswalk table.

Paper 3 - Context-complete interaction evidence

Single contribution: an observation and reporting standard that distinguishes traces from evidence.

Contents: closed-loop transition record, feasible/signaled/believed action sets, synchronization, uncertainty, provenance, and claim-dependent context.

Artifact: open schema, coding protocol, and validation examples.

Paper 4 - Interaction history as mediated learned state

Single contribution: a causal test of whether exposure history has effects beyond current measured knowledge, skill, expectations, habits, and external resources.

Contents: longitudinal design, mediator measurement, retention, forgetting, and competing models.

Artifact: longitudinal dataset and preregistered model-comparison suite.

Paper 5 - Typed equivalence and transfer

Single contribution: an empirical test of which equivalence types predict positive, negative, or zero transfer across representations and systems.

Contents: semantic, informational, task, computational, and interactional equivalence; similarity profiles; expert-novice differences.

Artifact: cross-system transfer benchmark.

Paper 6 - Coupled multi-timescale interaction model

Single contribution: a formal model linking joint state, interpretation, policy, transition, feedback, and learned state.

Contents: nested submodels, parameter identifiability, qualitative and quantitative predictions, and explicit omissions.

Artifact: executable reference model and simulation environment.

Paper 7 - Social and institutional extension

Single contribution: a test of when social roles, norms, language, division of labor, and institutional constraints add predictive or explanatory value beyond individual models.

Contents: collaborative tasks, breakdown and repair, distributed representations, multilevel models, and ethnographic validation.

Artifact: synchronized multi-actor corpus and coding framework.

Paper 8 - The decisive comparative test

Single contribution: determine whether any proposed interaction-specific mechanism adds stable explanatory power beyond integrated existing theories.

Contents: preregistered benchmark, held-out domains, model comparison, replication, and an explicit discipline decision.

Artifact: public benchmark, model leaderboard with uncertainty, and final boundary statement.

19. Formal argument

19.1 Argument against creating an independent discipline now

49. A new scientific discipline is justified only when it supports a progressive explanatory program that cannot be pursued adequately within existing disciplines.
50. The proposed domain - human interaction with computational processes - is coherent and implementation-independent when defined as reciprocal causal coupling over time.
51. Every proposed core phenomenon has an established explanatory counterpart: experience in learning and transfer; opportunity and constraint in ecological psychology, decision science, and human factors; representation and meaning in semiotics and cognitive science; context in situated action and anthropology; system coupling in cybernetics; collective organization in distributed cognition and activity theory; formal prediction in cognitive modeling and computational interaction.
52. The remaining weaknesses are primarily failures of integration, operational alignment, shared evidence standards, and cross-domain model comparison.
53. Integration failures do not by themselves establish a new mechanism, ontology, or natural kind.
54. Creating a new discipline before demonstrating conceptual irreducibility would duplicate terminology, fragment evidence, and weaken comparison with established baselines.
55. A focused research program within HCI can conduct the required ontology, measurement, longitudinal, and model-comparison work without foreclosing a later disciplinary claim.
56. Therefore, the creation of an independent discipline is not presently scientifically justified.

19.2 Strongest counterargument

The strongest positive argument is that no existing discipline treats temporally extended human-computational coupling - including representation, learning, system agency, social context, and cross-system transfer - as its central explanandum. HCI is broad and often design-led; cognitive science may treat technology as an environment rather than a co-adaptive agent; social theories often lack formal prediction; computational models often omit meaning and institutions. A dedicated discipline could force integration and cumulative standards.

19.3 Response

Centralization can improve coordination, but scientific necessity cannot be inferred from organizational convenience. The proposed integration can be tested as a research program. If it produces irreducible mechanisms and stable predictive laws, the evidence for a discipline will then exist. Naming the discipline first would reverse the required order: institution before necessity.

20. Final conclusion

Interaction does not currently justify an independent scientific discipline. It justifies a rigorous foundational research program within HCI, connected to cognitive science, human factors, learning science, and social theory.

The preferred theory is therefore not a new grand system. It is the smallest integration that permits decisive comparison:

- interaction as a coupled trajectory;

- action interpreted relative to feasible, signaled, and believed opportunities;
- representation separated from content and interpretation;
- history mediated through learned state and changed environments;
- equivalence specified by type;
- traces converted to evidence through an observation model;
- social and institutional variables included through mechanisms rather than demographic determinism;
- new constructs admitted only when they add replicated predictive power.

This conclusion should be reversed only by evidence that the integrated existing-theory baseline fails systematically and that a new interaction-specific mechanism succeeds across implementations. Until then, the scientifically conservative and more explanatory choice is integration rather than disciplinary proliferation.

21. Limitations

- This is a comprehensive boundary analysis, not an exhaustive systematic review of every subfield, language, or publication venue.
- The adjacent disciplines are internally diverse; summary boundaries identify dominant strengths and recurring limitations rather than unanimous positions.
- The formal model is intentionally generic. Without specific submodels and risky predictions it would be a notation, not an explanation.
- The report cannot empirically settle predictive increment; that requires the proposed benchmark program.
- Claims about universality are deliberately restricted to broad constraints and remain open to revision through disability, developmental, and cross-cultural research.
- The ontology prioritizes explanatory interoperability. It should not erase concepts needed for particular traditions, especially motive, power, identity, emotion, and institutional history.

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